Assignment 2 - OOD Detection: Deep Ensemble vs Autoencoder

Setup

In-distribution (ID): UC Merced test set, 315 aerial images, 21 classes. Out-of-distribution (OOD): FIDS30 fruit dataset, 971 images, 30 categories. Threshold at 95th percentile of ID scores targets ~5% false alarm rate (FAR).

Deep Ensemble

10 EfficientNet-B0 models, 5 epochs each, different random seeds. OOD score = variance of softmax outputs across models. High variance means models disagree, indicating a likely OOD sample.

Threshold: 0.006234 | Detection rate: 89.7% | False alarm rate: 3.8%

Nearly 90% of fruit images are correctly flagged while staying well below the 5% FAR target.

Convolutional Autoencoder

Encoder-decoder trained on UC Merced only (Adam lr=1e-3, MSE loss, 15 epochs). OOD score = per-sample reconstruction MSE. OOD images reconstruct poorly because the model has only seen aerial imagery.

Threshold: 0.012567 | Detection rate: 90.9% | False alarm rate: 3.8%

Comparison & Discussion

	Ensemble	Autoencoder
Detection rate:	89.7%	90.9%
False alarm rate:	3.8%	3.8%
Training cost:	10x models	1 model
Labels needed:	Yes	No

Both methods perform almost identically - kind of surprising. The autoencoder edges out slightly (90.9% vs 89.7%) while being cheaper and requiring no class labels at all. Real advantage: no labeling cost when scaling to new domains.

The ensemble is useful if you are already training classifiers and want OOD detection as a byproduct. But for this benchmark the autoencoder wins: simpler, label-free, marginally better. If the OOD domain were closer to ID data, the ensemble might win due to richer semantic uncertainty.